# Celso Manuel González Collado

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## SUMMARY

Physicist and computational scientist with experience in Python, Fortran, Linux, Shell Scripting, MATLAB, git, and high-performance computing (HPC), with a strong problem-solving mindset through analytical and critical thinking. Over the past years, I have developed numerical models, complex simulations, and data analysis workflows in advanced scientific environments. I have experience leading projects and collaborating in international teams across multiple disciplines.

## PROFESSIONAL EXPERIENCE

**PhD Researcher in Computational Modelling**

**Universidad Autónoma de Madrid (UAM)** 2021 – Present

- Development of cutting-edge simulations in optical and molecular physics, executed on world-class supercomputers and clusters like MareNostrum 5 (BSC) and the Scientific Computing Center at UAM.

- Strong use of Linux environments and communication protocols like ssh and scp, as well as essential HPC tools such as SLURM, module management (Lmod), parallelization libraries (MPI, OpenMP), following the best practices like using SCRATCH for heavy workloads.

- Development of scientific software in collaboration with Stanford University, including a 3 month research stay at Stanford (California, USA) working with physicists, chemists and software engineers.

- Programming in Python, MATLAB, Shell Scripting, and Fortran for data analysis, automation, modelling and code optimization for HPC. Use of Git for version control.

- Publication of scientific articles and oral presentations at international conferences in Vigo (Spain) and Washington (USA), where I was awarded by the American Physical Society for participation in APS DAMOP 2023. Experience in project management and planning.

- Teaching of multiple subjects in both Physics and Chemistry degrees at UAM, including Computational Tools for Chemists, Optics and Laboratory of Physical Chemistry.

## EDUCATION

**PhD in Theoretical Chemistry and Computational Modelling**

Universidad Autónoma de Madrid, June 2026

**Master's Degree in Theoretical Physics** – Score: 9.39/10

Universidad Complutense de Madrid, 2020

**Bachelor's Degree in Physics** – Score: 9.22/10

Universidad Complutense de Madrid, 2019

**Professional music degree** – Specialization in oboe

Conservatorio Profesional de Música Joaquín Turina (Madrid), 2015

## TECHNICAL SKILLS

**Programming Languages**

- Python (NumPy, SciPy, Pandas, Matplotlib)
- Shell Script
- Fortran
- MATLAB

**Technologies & Tools**

- Linux
- HPC (clusters, job scheduling, parallelization)
- Git
- LaTeX
- VSCode
- Microsoft Office (Word, Excel, PowerPoint)

**Areas of Expertise**

- Mathematical modelling
- Physical and chemical simulation
- Data processing and analysis
- Computational optimization
- Scientific software development

**Languages**

- Spanish: Native
- English: Advanced

**Other Skills**

- Driver's license and own vehicle

## SCIENTIFIC PUBLICATIONS

- J. Delgado, C. M. González-Collado, P. Decleva, A. Palacios & F. Martín, Physical Review A 111, 063107 (2025). DOI: 10.1103/qzg1-2f7n.
- C. M. González-Collado, L. Cattaneo, E. Plésiat, J. J. Omiste, J. Vos, J. González-Vázquez, P. Decleva, U. Keller, A. Palacios & F. Martín, Physical Review Research 6, 033342 (2024). DOI: 10.1103/PhysRevResearch.6.033342.
- V. Loriot, A. Boyer, S. Nandi, C. M. González-Collado, E. Plésiat, A. Marciniak, C. L. Garcia, Y. Hu, M. Lara-Astiaso, A. Palacios, P. Decleva, F. Martín & F. Lépine, Nature Physics 20, 765–769 (2024). DOI: 10.1038/s41567-024-02406-2.
- C. M. González-Collado, E. Plésiat, P. Decleva, A. Palacios & F. Martín, Physical Chemistry Chemical Physics (PCCP) 24, 7700 (2022). DOI: 10.1039/D2CP00627H.